EXPERIMENT NO: 14 **Date:** _____

NUMPY

AIM

Write a program to create a matrix using numpy and do the following operations: Perform matrix addition, subtraction and multiplication, Find transpose of matrix, Calculates the inverse of matrix, Calculates the determinant of matrix, Transforms a matrix into 1d array (flatten)

OBJECTIVES

- To utilize NumPy library for numerical array processing.
- To perform fundamental matrix operations (addition, subtraction, dot product).
- To compute linear algebra functions (transpose, inverse, determinant, flattening).

ALGORITHM

Step 1 :- Initialize matrix computation module using NumPy library for matrix arithmetic and linear algebra operations.

Step 2 :- Import numpy module under standard alias np to access array creation and matrix functions.

Step 3 :- Create two 2D arrays matrix1 and matrix2 with 2x2 integer values for calculations.

Step 4 :- Perform element wise addition, subtraction, and dot product multiplication using NumPy built in functions.

Step 5 :- Compute matrix transpose using np.transpose function to swap rows and columns of matrix1.

Step 6 :- Compute matrix inverse using np.linalg.inv and matrix determinant using np.linalg.det on matrix1.

Step 7 :- Transform 2D array matrix1 into 1D array flattened using matrix flatten method call.

Step 8 :- Print original matrices and calculated addition, subtraction, multiplication, transpose, inverse, determinant, and flattened outputs.

Step 9 :- Conclude NumPy matrix operations program execution and free array memory allocations from environment heap.

PROGRAM

```
import numpy as np

# Create matrices
matrix1 = np.array([[1, 2], [3, 4]])
matrix2 = np.array([[5, 6], [7, 8]])

# Perform matrix addition
addition = np.add(matrix1, matrix2)

```

```
# Perform matrix subtraction
subtraction = np.subtract(matrix1, matrix2)

# Perform matrix multiplication
multiplication = np.dot(matrix1, matrix2)

# Find transpose of matrix1
transpose = np.transpose(matrix1)

# Calculate the inverse of matrix1
inverse = np.linalg.inv(matrix1)

# Calculate the determinant of matrix1
determinant = np.linalg.det(matrix1)

# Transform matrix1 into a 1D array (flatten)
flattened = matrix1.flatten()

# Display results
print("Matrix 1:\n", matrix1)
print("\nMatrix 2:\n", matrix2)
print("\nMatrix Addition:\n", addition)
print("\nMatrix Subtraction:\n", subtraction)
print("\nMatrix Multiplication:\n", multiplication)
print("\nTranspose of Matrix 1:\n", transpose)
print("\nInverse of Matrix 1:\n", inverse)
print("\nDeterminant of Matrix 1:\n", determinant)
print("\nFlattened Matrix 1:\n", flattened)
```

SAMPLE OUTPUT

Terminal Output

```
Matrix 1:
[1 2

4
]


Matrix 2:
[5 6

8

]


Matrix Addition:
[ 6 8

12

]


Matrix Subtraction:
[-4 -4

-4
]


Matrix Multiplication:
[19 22

50









]


Transpose of Matrix 1:
[1 3

4
]


Inverse of Matrix 1:
[-2.  1.

-0.5
]


Determinant of Matrix 1:
-2.0


Flattened Matrix 1:
2 3 4
```

RESULT

The matrix operations program using NumPy library was executed and results verified.